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# Corporate Sustainability and Firm Valuation in the Global Oil and Gas Industry: The Moderating Role of External Sustainability Assurance

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**Abstract:** The relationship between corporate sustainability performance and firm valuation has attracted considerable scholarly and practical attention; however, empirical evidence remains inconclusive, particularly within environmentally sensitive and carbon-intensive industries. This study examines the association between corporate sustainability performance and firm valuation in the global oil and gas (O&G) sector and further investigates whether external sustainability assurance moderates this relationship. Grounded in Stakeholder Theory and Legitimacy Theory, an empirical analysis was conducted using a sample of 100 publicly listed O&G companies across multiple jurisdictions during the 2022–2023 period. Corporate sustainability performance was measured using Environmental, Social, and Governance (ESG) scores, while firm valuation was employed as the primary indicator of financial outcomes. In addition, the presence of independent third-party sustainability assurance was incorporated as a moderating variable to assess whether externally verified sustainability disclosures enhance the credibility and economic relevance of sustainability initiatives. The findings indicate that corporate sustainability performance is not significantly associated with firm valuation within the O&G industry. Furthermore, no significant moderating effect of external sustainability assurance was identified. These results suggest that sustainability-related activities and disclosures may not yet be perceived by investors as value-enhancing mechanisms in carbon-intensive sectors. It is also possible that stakeholders regard such initiatives as symbolic responses to legitimacy pressures rather than as substantive drivers of long-term economic performance. The findings challenge the widely accepted assumption that superior sustainability performance necessarily translates into improved market valuation and financial benefits. By providing evidence from a sector characterised by substantial environmental exposure, regulatory scrutiny, and stakeholder pressure, this study contributes to the growing literature on the economic consequences of corporate sustainability. The results further underscore the importance of developing industry-specific sustainability frameworks and assurance practices capable of strengthening stakeholder confidence and improving the integration of sustainability considerations into corporate value creation processes.

**Keywords:** Corporate Social Responsibility; Firm valuation; ESG performance; External sustainability assurance; Oil and gas industry; Stakeholder Theory; Legitimacy Theory

## 1. Introduction

Corporate Social Responsibility (CSR) has undergone a profound transformation over the past decades (Freeman & Hasnaoui, 2011), evolving from a largely philanthropic and voluntary practice into a strategic and institutionalised component of corporate governance. This shift has been driven by a combination of corporate scandals, environmental crises, and increasing stakeholder awareness regarding sustainability and ethical business conduct (Carroll, 2021; Noronha et al., 2013). The growing institutionalisation of CSR is evident in the widespread adoption of Environmental, Social, and Governance (ESG) frameworks (Carroll, 2008; Harjoto & Laksmana, 2018), which provide structured and measurable indicators of sustainability performance. Today, the vast majority of large multinational corporations disclose ESG-related information (KPMG, 2022), driven by regulatory developments, investor demand, and reputational considerations. Within this evolving landscape, firms operating in in controversial or carbon-intensive industries have emerged as frontrunners in CSR adoption. These firms face heightened scrutiny due to their environmental footprint and societal impact, which incentivises more extensive and detailed sustainability engagement (Al Farooque & Ahulu, 2017; De Villiers & Marques, 2016). However, such pressures may also encourage symbolic or strategic disclosure rather than substantive operational change, raising questions about the true effectiveness of CSR initiatives. Furthermore, balancing sustainability objectives with profitability remains a significant challenge, especially in industries characterised by high operational costs and environmental risks. Within this context, the oil and gas (O&G) industry provides a particularly relevant setting for examining CSR dynamics and their relationship with corporate financial performance. The industry is a major contributor to climate change and environmental degradation, accounting for a substantial share (62%) of global energy-related emissions (International Energy Agency, 2021). Its core activities—exploration, extraction, production, and distribution—generate significant environmental externalities and expose firms to considerable social and regulatory pressures. Moreover, O&G companies operate across diverse institutional and socio-cultural environments, often directly affecting local communities and multiple stakeholder groups. Controversial practices such as hydraulic fracturing (fracking) have further intensified public concern due to their adverse effects on soil and water resources (Brantley et al., 2014). Consequently, the sector is subject to intense scrutiny from governments, environmental organisations, and civil society actors seeking to mitigate its negative externalities (Boudet et al., 2014). Although the literature documents a growing integration of ESG considerations within the O&G industry, significant challenges remain. In particular, firms face difficulties in reconciling environmental performance with financial objectives (Harjoto & Laksmana, 2018), as well as in ensuring that CSR initiatives deliver meaningful and lasting benefits to local communities (Hurduzeu et al., 2022). This tension raises concerns that CSR practices in the sector may, in some cases, prioritise visibility and reputational gains over substantive engagement, potentially neglecting the most critical social and environmental issues (Raufflet et al., 2014). Despite increasing scholarly attention to the relationship between CSR and financial performance, empirical evidence remains fragmented (Coelho et al., 2023), with results varying due to differences in samples and institutional contexts. In this regard, recent literature has emphasized the need for industry-specific analyses, as the CSR–financial performance relationship appears to be highly context-dependent (Ye et al., 2021). Moreover, although recent literature has started to investigate the role of CSR assurance, limited attention has been devoted to carbon-intensive industries (Gallego-Álvarez & Pucheta-Martínez, 2022; Wang et al., 2016). Against this background, this research applies stakeholder and legitimacy theories to empirically investigate whether CSR performance influences firm valuation within the global O&G industry. Using a sample of the top 100 publicly listed O&G firms worldwide during the period 2022–2023, it also explores the moderating role of external assurance in this relationship, addressing the inconclusive findings reported in prior research (Wang et al., 2016). This study contributes to the literature in two main ways. First, it extends prior research on the CSR–financial performance nexus by providing industry-specific evidence that challenges the widely held assumption of a positive relationship between CSR engagement and financial outcomes. Second, it adds knowledge to the emerging research stream on the limitations and challenges associated with CSR disclosures in high-impact industries. In this regard, the study offers a more critical perspective on the role of credibility-enhancing mechanisms, showing that external assurance does not necessarily strengthen the financial relevance of CSR activities among global O&G companies. The paper proceeds as follows. The next section reviews the relevant literature and develops the research hypotheses. Section 3 outlines the research design and methodology, while Sections 4 and 5 present the empirical findings and discuss their implications. Finally, Section 6 concludes the study and provides recommendations for future research. Overall, the findings contribute to the ongoing debate on whether sustainability initiatives generate tangible economic benefits in industries characterised by significant environmental and social externalities.

## 2. Theoretical Framework and Hypotheses Development

CSR is increasingly recognised as a strategic organisational response to the challenges of sustainable development (Kaźmierczak, 2022). According to the World Business Council on Sustainable Development (Holme & Watts, 2000), CSR represents a management approach whereby firms contribute to sustainable economic development, integrating social and environmental concerns into their business operations and stakeholder interactions. Although the concept has historically been associated with philanthropy or narrowly defined social initiatives, it has progressively evolved into a more comprehensive and structured framework. In recent years, this evolution has been reflected in the growing prominence of ESG criteria, which provide measurable and standardised indicators of corporate sustainability performance. While CSR broadly captures a firm's commitment to responsible and sustainable practices, ESG translates these commitments into quantifiable metrics used by investors, regulators, and other stakeholders to assess firm performance (Kaźmierczak, 2022). This shift from qualitative CSR narratives to data-driven ESG evaluation has enhanced transparency and comparability, reinforcing the strategic importance of sustainability in corporate decision-making. Extant research documents that firms' CSR behaviours vary considerably between sensitive and non-sensitive industries (De Villiers et al., 2022). Firms operating in high-impact industries often face stronger pressure from regulators, investors, local communities, and environmental organisations to demonstrate responsible business conduct. As noted by García-Meca & Martínez-Ferrero (2021) sensitive-industry firms deliberately publish a good deal of data about their environmental performance in order to avoid incurring socio-political consequences. Additionally, the firms in sensitive sectors with stronger sustainability performance and disclosures have a greater Market Value (MV) than firms in non-sensitive industries.

In the 21st century, the O&G industry faces unprecedented challenges, primarily driven by climate change concerns and the global transition toward a low-carbon economy (Emeka-Okoli et al., 2024). Moreover, firms operating in this sector are particularly vulnerable to environmental and human rights controversies, given the significant social and ecological externalities associated with their operations (Hilson, 2012; Ruggie, 2013). In response to these pressures, O&G companies have increasingly sought to formalise their sustainability strategies and align their practices with evolving stakeholder expectations. A growing number of firms have adopted internationally recognised standards and initiatives to guide their CSR and sustainability efforts. Among these, the Global Reporting Initiative (GRI) has emerged as one of the most widely used frameworks, providing structured guidance on key areas such as environmental impacts, labour practices, human rights, and anti-corruption measures (Michalczuk & Konarzewska, 2020). In parallel, global sustainability initiatives and climate-related policy frameworks—such as the Paris Agreement, the United Nations Global Compact (UNGC) and the Carbon Disclosure Project (CDP)—have further encouraged firms to develop more comprehensive and forward-looking sustainability strategies, particularly in carbon-intensive industries. Although O&G companies are under increasing pressure to mitigate their environmental and social impacts, their CSR and sustainability practices often fall short of stakeholder expectations and require substantial improvement (Duttagupta et al., 2021). Persistent concerns regarding environmental degradation, community impacts, and the credibility of corporate commitments suggest that, in this sector, CSR engagement remains an evolving and contested process (Raufflet et al., 2014).

From a theoretical standpoint, the rationale underlying the "business case" for CSR is primarily associated with Stakeholder Theory and Legitimacy Theory. Stakeholder Theory argues that firms generate long-term value by effectively addressing the interests of multiple stakeholder groups, implying that CSR initiatives may improve corporate performance through reputational gains, stronger stakeholder relationships, and enhanced access to resources. Conversely, Legitimacy Theory suggests that organisations implement CSR practices to conform to societal norms and preserve their "social licence to operate," especially in industries subject to intense public attention and regulatory scrutiny. In this study, these two theoretical perspectives provide the basis for examining both the relationship between CSR performance and financial outcomes and the potential moderating effect of external assurance. By combining these approaches, the analysis seeks to determine whether CSR activities in the O&G industry create tangible economic benefits or mainly serve as mechanisms for maintaining organisational legitimacy.

### 2.1 Corporate Social Responsibility and Financial Performance

Stakeholder theory (Freeman, 1984) posits that a firm's long-term success is contingent upon its ability to balance and satisfy the needs of diverse groups—including investors, employees, and the public—rather than focusing solely on shareholder returns. According to this view, CSR disclosure serves as a mechanism to manage these interdependent relationships and secure the support of critical stakeholders. Supporters of this view argue that superior CSR performance enhances a firm's reputation, facilitates the attraction of high-quality talent, and reduces the cost of capital, ultimately translating into improved financial performance. Empirically, numerous studies have identified a positive correlation between CSR activities and firm's financial performance (Beck et al., 2018), suggesting that socially responsible behaviour is rewarded by investors. For instance, Al-Dah et al. (2018) and Jizi et al. (2014) reported a positive relationship between CSR performance and financial outcomes, noting that more profitable firms have greater resources to invest in social initiatives, which can, in turn, enhance their reputation and financial standing. This positive feedback loop has been observed in several countries, with studies from India (Oware & Mallikarjunappa, 2022) demonstrating a positive correlation between CSR disclosures and improved financial performance. While a consistent stream of literature suggests that CSR engagement may have material implications for corporate financial performance Ullah, other studies that explore this relationship have reported mixed results (Kahloul et al., 2022; Mukherjee & Nuñez, 2019). In this vein, the study of Han et al. (2016) found that CSR expenditure leads to improved financial performance only when it surpasses a certain threshold. Singhania et al. (2024) further document revealed a positive link between CSR reporting and accounting-based financial performance, but a negative link with market-based performance. These contradictory results indicates that the relationship between CSR and financial performance is complex rather than linear, as it may depend on several contextual elements, including industry-specific characteristics that the link between. From a legitimacy theory perspective, companies adopt CSR practices to satisfy societal expectations and preserve their legitimacy within the broader social environment. When firms meet these standards, they are often rewarded with enhanced reputation and better economic results. However, the "win-win" paradigm is frequently contested in the context of high-emitters companies. Indeed, the demand for more CSR is not uniform across industries or across firms within industries because firms and industries vary in the amount of attention they attract. Firms operating in environmentally intensive sectors, such as O&G, are subject to higher levels of public scrutiny and legitimacy pressures compared to less visible industries. As a result, CSR engagement in these contexts is often driven by the need to maintain legitimacy rather than by intrinsic commitments to sustainability. In the O&G sector, CSR initiatives are frequently interpreted through the lens of instrumental legitimacy, whereby firms adopt ESG frameworks and sustainability disclosures primarily to signal alignment with societal expectations. Critics argue that such practices may lack substantive commitment, raising concerns about symbolic management and "greenwashing." In this context, CSR practices may function as a strategic tool aimed at preserving legitimacy while minimising the costs associated with deeper organisational transformation (Raufflet et al., 2014). In fact, the high costs associated with regulatory compliance, operational restructuring and environmental stewardship in this sector can act as a financial burden, potentially offsetting the potential economic benefits of CSR engagement. Consistent with this view, some strands of the literature suggest that in high-impact industries, environmental disclosures and CSR investments may not translate into improved financial performance and, in certain cases, may even adversely affect accounting-based measures. For example, Hurduzeu et al. (2022) and Matuszak & Różańska (2017) observed weak correlations between CSR engagement and financial performance in energy-related sectors. Similarly, Strouhal et al. (2015) suggest that while CSR initiatives are increasingly adopted in O&G sectors, they might not translate into immediate financial benefits in the O&G sector. Taken together, these findings highlight the need for further investigation into the CSR–financial performance relationship, particularly within environmentally sensitive industries. Building on this discussion, the present study advances the following hypothesis:

**Hypothesis 1:** *There is a positive relationship between CSR performance and firms' financial performance.*

## 2.2 Moderation of Third-Party Assurance

Despite the rapid expansion of CSR reporting, concerns persist regarding the quality and consistency of such disclosures (Dhaliwal et al., 2012). In fact, CSR reporting remains largely voluntary (De Villiers & Marques, 2016), resulting in fragmented and heterogeneous disclosure practices that hinder comparability across firms. This "credibility gap" makes it difficult for stakeholders, particularly investors, to distinguish between substantive sustainability efforts and symbolic practices often associated with "greenwashing." In this context, external assurance has emerged as a key mechanism to enhance the value relevance of CSR disclosures. According to the IFRS, assurance services provide an independent evaluation of an organisation's governance, risk management, and control processes. Applied to non-financial information, external assurance thereby improves the integrity and reliability of the CSR practices communicated to stakeholders. From a theoretical standpoint, external assurance functions can be viewed as a legitimation process that enhances corporate reputation and reinforces stakeholder trust (Amran et al., 2024; Simnett et al., 2009). It reduces information asymmetry between firms and external stakeholders, enhancing the value relevance of CSR activities and boosting investors' confidence to invest in the company (Xie et al., 2022). As a result, companies with significant environmental and social consequences are likely to seek external assurance for demonstrating their social responsibility for maintaining their reputation and revenues. Empirical evidence largely supports the role of external assurance for improving CSR and financial performance, although findings vary across different institutional contexts. Gallego-Álvarez & Pucheta-Martínez (2022) provide international evidence that assurance positively moderates the relationship between CSR disclosure and firm performance, suggesting that assured information is perceived as more decision-useful by investors. Similarly, Kim et al. (2019) find that firms issuing assured sustainability reports exhibit superior financial performance compared to non-assured counterparts, while Uyar et al. (2023) document that external verification helps firms achieve greater CSR performance. Casey & Grenier (2015) further demonstrate that assurance reduces the cost of equity capital and improves the accuracy of analyst forecasts, particularly when provided by reputable accounting firms. However, other studies highlight that the impact of CSR assurance on firm's performance is highly contingent upon geographical and market-specific factors (Oware & Mallikarjunappa, 2019). This is because the sustainability assurance process for multinational firms is characterised by a higher degree of variability across countries as well as among firms (Farooq & de Villiers, 2019). Given the above arguments, we expect that third-party assurance improves the credibility of CSR disclosures, leading to better financial outcomes. Accordingly, the following hypothesis was proposed:

**Hypothesis 2:** *CSR assurance moderates (strengthens) the positive relationship between CSR performance and financial performance.*

## 3. Research Design

### 3.1 Sample selection and Corporate Social Responsibility Performance

This study examines the O&G sector to provide industry-specific evidence on the economic and financial effects of CSR engagement, using a sample of the top 100 publicly listed companies worldwide based on market capitalisation. The sample was constructed using Refinitiv Eikon by applying a screening procedure based on the Thomson Reuters Business Classification (TRBC). Companies classified within the O&G sector were identified and ranked according to market capitalisation, and the top 100 firms were selected to ensure the inclusion of the largest and most financially robust companies in the industry. The dataset was further refined by retaining only firms with available ESG and financial data over a two-year period (2022–2023). This timeframe reflects the post-pandemic market environment and a period of increasing standardisation and regulatory attention in ESG reporting, characterised by the introduction of new global disclosure requirements and the expansion of climate-related reporting obligations across jurisdictions (Christensen et al., 2021; Hummel & Jobst, 2024). The final sample consists of 96 firms, resulting in a cross-country dataset of 192 firm-year observations that captures firms operating under diverse institutional and regulatory environments. CSR performance is proxied by the ESG score provided by Refinitiv Eikon. ESG scores are constructed across three primary pillars—ESG—each comprising multiple sub-categories that capture different dimensions of corporate sustainability. Following the methodology outlined by Kim et al. (2019), ESG ratings, originally expressed on a letter scale ranging from A+ to D−, are converted into a numerical scale to facilitate quantitative analysis. In line with Refinitiv's scoring guidelines, these ratings are transformed into values ranging from 0.083 (D−) to 1 (A+), reflecting the full spectrum of twelve rating categories. This standardisation allows for meaningful comparison across firms and enables the application of regression techniques for empirical analysis. Table 1 presents the sample distribution. The number of firms disclosing CSR information remains stable across the two years, with 96 out of 100 companies reporting ESG data in both 2022 and 2023. The mean CSR score shows a slight increase from 0.679 in 2022 to 0.682 in 2023, indicating a modest improvement in overall CSR performance. However, the proportion of firms obtaining external assurance declines marginally from 72.92% to 70.83%, suggesting a slight reduction in audit coverage over the period.

**Table 1.** Sample distribution

| Year | No. of Companies Observed | No. of Companies with CSR Report | Mean CSR Score | % of Audit |
|---|---|---|---|---|
| 2022 | 100 | 96 | 0.679 | 72.92% |
| 2023 | 100 | 96 | 0.682 | 70.83% |

Note: CSR = Corporate Social Responsibility.

### 3.2 Variables

3.2.1 Dependent variables

Firm financial performance is measured using two market-based indicators, consistent with prior literature (Kim et al., 2019): MV and Tobin's Q (TOBINQ). These measures capture investors' expectations regarding firms' future profitability and growth prospects. The first measure, MV, is defined as the ratio of the firm's MV—calculated as the stock price multiplied by the number of outstanding shares—to the book value of total assets net of cash and marketable securities. Formally, for firm i in year t, MV is computed as follows:

$$MV_{i,t} = \text{(the stock price of the firm * number of shares outstanding)/(book value of total assets − cash and marketable securities)}$$

The second measure, TOBINQ, captures the market valuation of a firm relative to its asset base. It reflects investors' expectations regarding the firm's ability to generate future returns. Following Lee & Grewal (2004), TOBINQ is computed as:

$$TOBINQ_{i,t} = \text{(MV of equity + preferred stock + short-term liabilities + book value of long-term debt)/(book value of total assets)}$$

3.2.2 Control and independent variables

To isolate the relationship between CSR performance and financial performance, several control variables commonly used in the literature are included (Kim et al., 2019). Return on Assets (ROA) is measured as the ratio of income before extraordinary items to total assets and serves as a proxy for firm profitability and operational efficiency. Firm size (SIZE) is measured as the natural logarithm of total assets, capturing scale effects and differences in firm visibility. Leverage (LEV) is calculated as the ratio of total debt (short-term and long-term) to equity, reflecting the firm's financial structure. The book-to-market ratio (BM), defined as the ratio of the book value of equity to its MV, is included as a proxy for firm valuation and growth opportunities. Finally, firm growth (GROWTH) is measured as the annual sales growth rate, calculated as the change in sales divided by lagged sales. CSR performance is proxied by the ESG score obtained from Refinitiv Eikon, as described in Section 3.1. To examine the moderating role of external assurance, an indicator variable (AUDIT) is introduced. AUDIT is a binary variable equal to 1 if a firm's CSR report is externally assured by a professional third party, and 0 otherwise. Additionally, an interaction term (CSR × AUDIT) is constructed to capture the moderating effect of external assurance on the relationship between CSR performance and financial performance. In our analysis, year and industry fixed effects were not included in the regression models for several reasons. First, the analysis covers a limited time horizon (2022–2023), reducing the likelihood of substantial time-specific variation affecting the relationships tested. Second, although the sampled firms belong to different sub-industries, they exhibit relatively similar CSR practices and financial characteristics, reducing the need for industry-specific controls. In this regard, preliminary descriptive and correlation analyses did not reveal substantial temporal or sub-industry heterogeneity. Moreover, omitting these dummies reduces the risk of multicollinearity and overfitting issues that may arise in limited sample settings (Canela et al., 2019).

### 3.3 Regression Models

To empirically investigate the relationship between CSR performance and financial performance, we employ Ordinary Least Squares (OLS) regression analysis. The use of panel data allows us to control for unobserved heterogeneity among firms that remains constant over time, thereby mitigating potential omitted variable bias (BIBLO). Specifically, two distinct regression models are used for investigating the impact of CSR on two different financial performance measures: MV and TOBINQ. The OLS regression framework was utilized for both models, ensuring robust analysis and interpretation.

Model 1: MV as the Dependent Variable

This model examines the direct effect of CSR on MV, while controlling for other variables such as ROA, SIZE, LEV, BM, and GROWTH. The functional form is:

$$MV_{i,t} = \alpha 0 + \alpha 1\ CSR_{i,t} + \alpha 2\ ROA_{i,t} + \alpha 3\ SIZE_{i,t} + \alpha 4\ LEV_{i,t} + \alpha 5\ BM_{i,t} + \alpha 6\ GROWTH_{i,t} + \epsilon_{i,t}$$

Model 2: TOBINQ as the Dependent Variable

In this model, TOBINQ is used as the dependent variable to capture a market-based valuation of firm performance. The regression equation is similar to Model 1, with TOBINQ replacing MV as the dependent variable:

$$TOBINQ_{i,t} = \beta 0 + \beta 1\ CSR_{i,t} + \beta 2\ ROA_{i,t} + \beta 3\ SIZE_{i,t} + \beta 4\ LEV_{i,t} + \beta 5\ BM_{i,t} + \beta 6\ GROWTH_{i,t} + \epsilon_{i,t}$$

Model 3: Interaction Effects with Audit

To explore the moderating role of external audits on the relationship between CSR and financial performance, an interaction term between CSR and AUDIT is included. Separate regressions for MV and TOBINQ are specified as follows:

$$MV_{i,t} = \gamma 0 + \gamma 1\ CSR_{i,t} + \gamma 2\ AUDIT_{i,t} + \gamma 3\ CSR \times AUDIT_{i,t} + \gamma 4\ ROA_{i,t} + \gamma 5\ SIZE_{i,t} + \gamma 6\ LEV_{i,t} + \gamma 7\ BM_{i,t} + \gamma 8\ GROWTH_{i,t} + \epsilon_{i,t}$$

$$TOBINQ_{i,t} = \delta 0 + \delta 1\ CSR_{i,t} + \delta 2\ AUDIT_{i,t} + \delta 3\ CSR \times AUDIT_{i,t} + \delta 4\ ROA_{i,t} + \delta 5\ SIZE_{i,t} + \delta 6\ LEV_{i,t} + \delta 7\ BM_{i,t} + \delta 8\ GROWTH_{i,t} + \epsilon_{i,t}$$

where,

$AUDIT_{i,t}$ is a binary variable indicating whether the firm's CSR report is externally audited,

$CSR \times AUDIT_{i,t}$ represents the interaction term between CSR and external auditing.

Taken together, these models allow to analyze the impact of CSR on distinct measures of financial performance, reflecting different dimensions of firm value. By introducing the audit variable and its interaction with CSR, it can be examined whether external auditing strengthens the CSR-financial performance relationship, providing insights into the potential moderating effect of audit practices.

## 4. Results

### 4.1 Descriptive Statistics

Table 2 reports the descriptive statistics for CSR and its components over the period 2022–2023. Overall, the results indicate a modest improvement in aggregate CSR performance, with the mean CSR score increasing slightly from 0.679 to 0.682. This upward movement, although limited in magnitude, suggests a gradual strengthening of sustainability practices across the sample. At the same time, the standard deviation increased marginally (from 0.144 to 0.148), indicating a slight rise in dispersion and, therefore, growing heterogeneity in CSR performance among firms. Examining the individual ESG dimensions provides additional insights. Environmental (E) scores show a small increase in the mean (from 0.668 to 0.671), accompanied by a reduction in standard deviation (from 0.189 to 0.184), suggesting a more consistent alignment of firms toward environmental practices. Similarly, Social (S) scores exhibit both an increase in the mean (from 0.705 to 0.712) and a decline in variability (standard deviation decreasing from 0.172 to 0.163), indicating a convergence toward improved social performance. In contrast, Governance (G) scores display a slight deterioration, with the mean declining from 0.666 to 0.662 and the standard deviation increasing from 0.226 to 0.235. This pattern suggests greater dispersion and inconsistency in governance practices, which may reflect differences in internal control systems, board structures, or regulatory environments across firms.

**Table 2.** Descriptive statistics

| Variable | 2022 | | | | 2023 | | | |
|---|---|---|---|---|---|---|---|---|
| | **Obs.** | **Mean** | **Median** | **SD** | **Obs.** | **Mean** | **Median** | **SD** |
| E | 96 | 0.668 | 0.666 | 0.189 | 96 | 0.671 | 0.666 | 0.184 |
| S | 96 | 0.705 | 0.750 | 0.172 | 96 | 0.712 | 0.750 | 0.163 |
| G | 96 | 0.666 | 0.750 | 0.226 | 96 | 0.662 | 0.750 | 0.235 |
| CSR Score | 96 | 0.679 | 0.694 | 0.144 | 96 | 0.682 | 0.694 | 0.148 |

Note: E = Environmental; S = Social; G = Governance; CSR = Corporate Social Responsibility; Obs. = Observations; SD = Standard Deviation.

Notably, the median values remain unchanged across both years for all ESG components and the overall CSR score, indicating that central tendencies are stable, and that the observed changes are driven primarily by movements in the distribution tails rather than shifts in the typical firm. Overall, these findings highlight a gradual improvement in environmental and social dimensions, coupled with increasing variability in governance practices, which contributes to the slight increase in overall CSR dispersion.

Figure 1 illustrates the distribution of CSR scores across firms. The distribution is relatively wide, with most observations concentrated between 0.50 and 0.92, indicating substantial variation in CSR engagement levels within the sample. A clear clustering emerges around mid-range values (approximately 0.69–0.75), suggesting that the majority of firms adopt moderate CSR practices rather than extreme positions. High CSR scores (above 0.80) are comparatively rare, indicating that only a limited number of firms achieve advanced sustainability performance. Conversely, the presence of lower scores (particularly between 0.31 and 0.44) highlights that a subset of firms still exhibits relatively weak CSR engagement. This distribution confirms the existence of significant heterogeneity in CSR practices, which is consistent with the sector-specific challenges and varying levels of stakeholder pressure discussed in the literature.

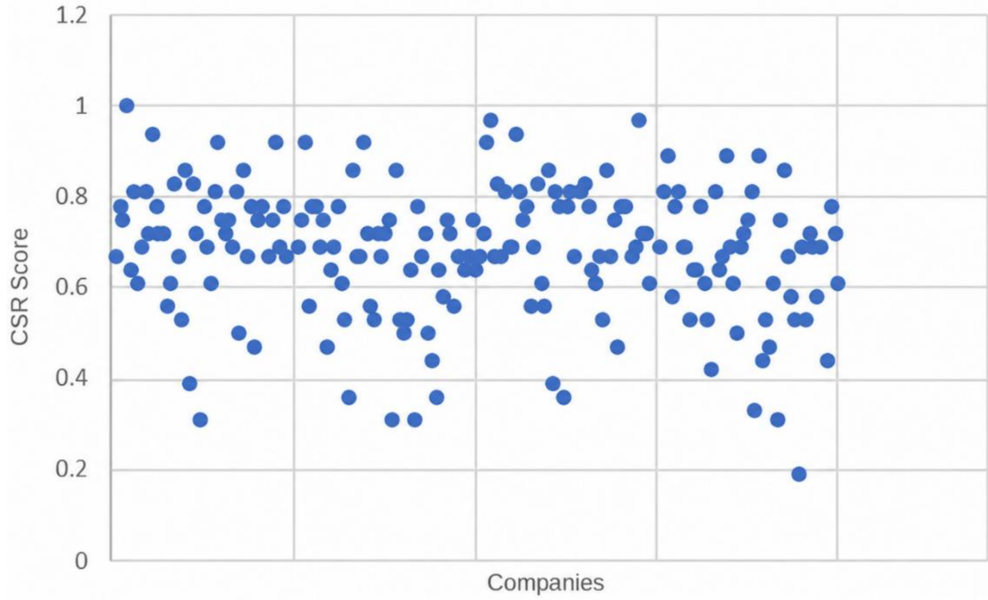


**Figure 1.** Corporate Social Responsibility (CSR) score

Table 3 reports the Pearson correlation coefficients among the variables. The results indicate weak and predominantly insignificant relationships between CSR performance and financial performance proxies. Specifically, CSR is negatively but weakly correlated with MV (−0.080) and TOBINQ (−0.076), suggesting the absence of a meaningful association between CSR engagement and market-based valuation. Similarly, the correlation between CSR and ROA is slightly negative (−0.088), indicating no clear link between CSR performance and accounting-based profitability. Taken together, these findings provide preliminary evidence that does not support the conventional expectation of a positive CSR–financial performance relationship within this sample. With respect to firm characteristics, CSR is weakly negatively associated with LEV (−0.071), suggesting that firms with higher CSR engagement may rely marginally less on debt financing. A more pronounced and statistically significant negative correlation is observed between CSR and the book-to-market ratio (−0.183, $p < 0.05$), indicating that firms with stronger CSR performance tend to exhibit higher market valuation relative to their book value, potentially reflecting reputational or intangible benefits. CSR is also negatively correlated with growth (−0.136), suggesting that higher CSR engagement does not necessarily coincide with stronger short-term sales growth. In contrast, CSR is positively and significantly associated with SIZE (0.191, $p < 0.01$), indicating that larger firms are more likely to engage in CSR activities, possibly due to greater resource availability and visibility.

**Table 3.** Correlation matrix of key variables

| Variable | CSR | MV | TOBINQ | LEV | ROA | BM | GROWTH | SIZE |
|---|---|---|---|---|---|---|---|---|
| CSR | 1 | | | | | | | |
| MV | -0.08 | 1 | | | | | | |
| TOBINQ | -0.076 | 0.999*** | 1 | | | | | |
| LEV | -0.071 | -0.018 | -0.017 | 1 | | | | |
| ROA | -0.088 | 0.067 | 0.057 | -0.079 | 1 | | | |
| BM | -0.183* | -0.032 | -0.036 | -0.022 | -0.262** | 1 | | |
| GROWTH | -0.136 | 0.117 | 0.113 | -0.032 | 0.280** | -0.127 | 1 | |
| SIZE | 0.191** | -0.053 | -0.062 | -0.013 | -0.01 | -0.031 | -0.084 | 1 |

Note: CSR = Corporate Social Responsibility; MV = Market Value; TOBINQ = Tobin's Q; LEV = Leverage; ROA = Return on Assets; BM = Book-to-Market Ratio; GROWTH = Growth Rate; SIZE = Firm size. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

As expected, MV and TOBINQ exhibit an almost perfect correlation (0.999, $p < 0.001$), reflecting their conceptual similarity. To mitigate potential multicollinearity concerns, these variables are included in separate regression models. Among control variables, ROA is positively correlated with growth (0.280, $p < 0.01$) and negatively associated with the book-to-market ratio (−0.262, $p < 0.01$), consistent with the expectation that more profitable firms tend to experience higher growth and higher market valuation.

Overall, the correlation analysis indicates that CSR performance is not strongly associated with financial performance measures in this sample. The weak and predominantly negative correlations suggest that the relationship between CSR and firm performance is likely to be complex, non-linear, and influenced by additional moderating factors, such as industry characteristics and institutional context. These preliminary findings underscore the need for multivariate analysis to better capture the underlying dynamics and to control for confounding factors. Accordingly, the next section presents the regression results to further investigate the CSR–financial performance relationship and the potential moderating role of external assurance.

## 4.2 Regression Results

4.2.1 Corporate Social Responsibility and financial performance

To test Hypothesis 1, two alternative proxies for financial performance—MV in Model 1 and TOBINQ in Model 2—are used as dependent variables. CSR performance, measured through ESG scores, is included as the main explanatory variable, while a set of firm-specific controls—ROA, SIZE, LEV, BM, and GROWTH—are incorporated to account for heterogeneity across firms. Table 4 reports the regression results for both model specifications. Overall, the findings indicate weak explanatory power and a lack of statistical significance across models, suggesting that CSR performance does not exert a measurable impact on firm valuation in the O&G sector.

**Table 4.** Regression results for Model 1 and 2

| Variable | Model 1 | | | | Model 2 | | | |
|---|---|---|---|---|---|---|---|---|
| | Coeff. | Std. Error | *t*-Value | Sig. | Coeff. | Std. Error | *t*-Value | Sig. |
| Constant | 3.988** | 1.846 | 2.160 | 0.032 | 4.386** | 1.825 | 2.404 | 0.017 |
| CSR | -3.026 | 2.307 | -1.311 | 0.191 | -2.899 | 2.280 | -1.271 | 0.205 |
| ROA | -1.084 | 5.925 | -0.183 | 0.855 | -1.958 | 5.856 | -0.334 | 0.738 |
| SIZE | 2.158E-7 | 0.000 | 0.057 | 0.954 | -2.816E-7 | 0.000 | -0.076 | 0.904 |
| LEV | -0.054 | 0.180 | -0.303 | 0.762 | -0.053 | 0.178 | -0.301 | 0.764 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| BM | -1.033** | 0.467 | -2.210 | 0.028 | -1.040** | 0.462 | -2.252 | 0.025 |
| GROWTH | 1.551 | 0.963 | 1.610 | 0.109 | 1.506 | 0.952 | 1.582 | 0.115 |
| *N. Obs.* | | 192 | | | | 192 | | |
| *N. Groups* | | 96 | | | | 96 | | |
| $R^2$ | | 0.044 | | | | 0.044 | | |
| Adjusted $R^2$ | | 0.014 | | | | 0.013 | | |
| F-statistic | | 1.448 | | | | 1.439 | | |
| Sign (*p*-value) | | 0.198 | | | | 0.202 | | |

Note: CSR (Corporate Social Responsibility) = ESG score from Thomson Reuters*;* ROA (Return on asset) = the ratio of income before extraordinary items to total assets; SIZE (Firm size) = the natural logarithm of total assets; LEV(Leverage) = the ratio of debt to equity; BM (Book-to-Market Ratio) = the ratio of book value to market value of equity; GROWTH (Growth Rate) = the ratio of the change in sales to lagged sales; Obs. = Observations; Coeff. = Coefficient; Std. Error = Standard Error; Sig. = Significance. ** indicates statistical significance at the 5% level ($p < 0.05$).

Specifically, the coefficient on CSR is negative and statistically insignificant in both models (MV: $\beta = -3.026$, $p = 0.191$; TOBINQ: $\beta = -2.899$, $p = 0.205$), indicating that higher CSR performance is not associated with improved financial outcomes. These results are consistent across both market-based measures, providing robust evidence against the hypothesised positive relationship. Among the control variables, only the BM exhibits a statistically significant negative relationship with financial performance (MV: $\beta = -1.033$, $p < 0.05$; TOBINQ: $\beta = -1.040$, $p < 0.05$), suggesting that firms with lower BM ratios (i.e., higher market valuation relative to book value) tend to achieve higher market-based performance. All other control variables—including ROA, SIZE, LEV, and GROWTH—do not display statistically significant effects in either specification. The overall model fit is weak, with both regressions yielding low R-squared values ($R^2 = 0.044$), indicating that only a small proportion of the variation in financial performance is explained by the included variables. Furthermore, the F-statistics are not statistically significant ($p > 0.10$), suggesting that the models lack joint explanatory power. Taken together, these findings provide no empirical support for Hypothesis 1, indicating that CSR performance does not significantly influence firm financial performance—measured by either MV or TOBINQ—in the O&G sector. This evidence reinforces the view that the CSR–financial performance relationship is context-dependent and may be weakened in environmentally sensitive industries characterised by high regulatory costs and legitimacy pressures.

4.2.2 Corporate Social Responsibility assurance and financial performance

Table 5 reports the regression results testing Hypothesis 2, which examines whether external assurance moderates the relationship between CSR performance and firm financial performance. The models include both MV and TOBINQ as dependent variables, while incorporating CSR performance, AUDIT, and their interaction term (CSR×AUDIT), alongside standard firm-level control variables.

**Table 5.** Regression results for Model 3

| **Variable** | **Model 3 (Dependent Variable = MVs)** | | | | **Model 3 (Dependent Variable = TOBIN)** | | | |
|---|---|---|---|---|---|---|---|---|
| | **Coeff.** | **Std. Error** | ***t*-Value** | **Sig.** | **Coeff.** | **Std. Error** | ***t*-Value** | **Sig.** |
| Constant | 5.544 | 2.938 | 1.887 | 0.061 | 6.025 | 2.904 | 2.075 | 0.039 |
| CSR | -3.826 | 4.655 | -0.822 | 0.412 | -3.843 | 4.600 | -0.835 | 0.405 |
| AUDIT | -3.077 | 3.945 | -0.780 | 0.436 | -3.086 | 3.898 | -0.792 | 0.430 |
| CSR×AUDIT | 2.863 | 6.031 | 0.475 | 0.636 | 2.915 | 5.959 | 0.489 | 0.625 |
| ROA | -1.775 | 6.219 | -0.285 | 0.776 | -2.711 | 6.145 | -0.441 | 0.660 |
| SIZE | 3.916E-7 | 0.000 | 0.099 | 0.921 | -1.188E-7 | 0.000 | -0.030 | 0.976 |
| LEV | -0.073 | 0.184 | -0.399 | 0.690 | -0.073 | 0.182 | -0.403 | 0.688 |
| BM | -1.269 | 0.584 | -2.173 | 0.031 | -1.272 | 0.577 | -2.203 | 0.029 |
| GROWTH | 1.483 | 0.981 | 1.512 | 0.132 | 1.441 | 0.969 | 1.487 | 0.139 |
| $R^2$ | | 0.058 | | | | 0.058 | | |
| Adjusted $R^2$ | | 0.016 | | | | 0.016 | | |
| F-statistic | | 1.439 | | | | 1.387 | | |
| Sign (*p*-value) | | 0.202 | | | | 0.205 | | |

Note: CSR = ESG score from Thomson Reuters*;* AUDIT = Dummy variable equal to 1 if the firm's CSR report is audited by an external professional and 0 otherwiseis; CSR*AUDIT = interaction variable; ROA (Return on asset) = the ratio of income before extraordinary items to total assets; SIZE (Firm size) = the natural logarithm of total assets; LEV(Leverage) = the ratio of debt to equity; BM (Book-to-Market Ratio) = the ratio of book value to market value of equity; GROWTH (Growth Rate) = the ratio of the change in sales to lagged sales; Coeff. = Coefficient; Std. Error = Standard Error; Sig. = Significance.

Overall, the results provide no evidence supporting the moderating role of CSR assurance. The interaction term (CSR × AUDIT) is positive in both specifications (MV: $\beta = 2.863$; TOBINQ: $\beta = 2.915$), but statistically insignificant ($p > 0.10$), indicating that external assurance does not strengthen the relationship between CSR performance and financial outcomes. This finding suggests that the presence of third-party verification does not enhance the market's valuation of CSR activities in the O&G sector. Moreover, the direct effect of CSR remains negative and statistically insignificant in both models (MV: $\beta = -3.826$, p = 0.412; TOBINQ: $\beta = -3.843$, $p$ = 0.405), consistent with the results reported for Hypothesis 1. Similarly, the coefficient on AUDIT is negative and insignificant across both specifications, indicating that the adoption of external assurance practices does not independently influence firm valuation. The explanatory power of the models remains limited, with $R^2$ values of 0.058 and non-significant F-statistics ($p > 0.10$), suggesting that the included variables explain only a small proportion of the variation in financial performance. Among the control variables, only the BM is statistically significant and negatively associated with both MV and TOBINQ ($p < 0.05$), indicating that firms with lower BM ratios tend to exhibit higher market valuation. All other controls—ROA, SIZE, LEV, and GROWTH—do not display significant effects.

## 5. Discussion

The empirical findings of this study consistently indicate that CSR performance does not exert a statistically significant effect on firm financial performance, as measured by both MV and TOBINQ, across all model specifications. The coefficients associated with CSR are negative and insignificant in Models 1 and 2, and this pattern persists in Model 3, where the inclusion of CSR assurance (AUDIT) and its interaction term (CSR × AUDIT) does not alter the results. Notably, neither the direct effect of CSR nor the moderating role of external assurance is statistically significant, suggesting that even credibility-enhancing mechanisms fail to strengthen the CSR–financial performance nexus in the O&G sector. These findings provide no support for hypotheses 1 and 2 and challenge the predictions of Stakeholder Theory, which posits that CSR engagement should enhance firm value through improved reputation, stakeholder relations, and resource access (Al-Dah et al., 2018; Jizi et al., 2014). Instead, the results align more closely with Legitimacy Theory, indicating that in environmentally sensitive and highly scrutinised industries such as O&G, CSR practices may function primarily as tools for maintaining legitimacy rather than generating economic returns. The lack of a significant moderating effect of assurance further suggests that external verification may be perceived by investors as symbolic rather than substantive, particularly in a context often associated with greenwashing concerns (Cho et al., 2019; Raufflet et al., 2014). These findings are consistent with a growing body of literature highlighting the context-dependent nature of the CSR–financial performance relationship, with several studies reporting weak or inconclusive results in energy and high-impact sectors (Hurduzeu et al., 2022; Matuszak & Różańska, 2017; Strouhal et al., 2015), as well as divergent effects depending on the performance measure employed (Singhania et al., 2024). Several explanations may account for these outcomes. First, the substantial costs associated with environmental compliance, sustainability investments, and operational restructuring in the O&G industry may offset potential financial gains from CSR activities. Second, persistent investor scepticism toward CSR disclosures—particularly in sectors prone to reputational risk—may limit the extent to which such information is incorporated into firm valuation. Specifically, in carbon-intensive industries, investors may perceive CSR activities and external assurance practices as insufficient to offset the long-term environmental risks, transition uncertainties, and regulatory pressures associated with fossil fuel-based business models (García-Amate et al., 2023). Consequently, greater emphasis may continue to be placed on traditional financial and operational indicators—such as profitability, cash flows, production capacity, and energy price dynamics—rather than on sustainability disclosures when assessing firm value. Furthermore, the absence of a significant market response may indicate that ESG engagement in the O&G sector is perceived as having limited short-term economic relevance, particularly when sustainability initiatives are not accompanied by substantial changes in firms' core business models or decarbonisation strategies. This interpretation is consistent with prior studies suggesting that, although CSR activities may contribute to reputational management and stakeholder communication, investors may not interpret them as reliable indicators of future competitive advantage or long-term value creation in environmentally sensitive industries (Cho et al., 2019; Raufflet et al., 2014). At the same time, it is also possible that the economic benefits associated with CSR initiatives materialise over a longer time horizon, whereas market-based indicators such as TOBINQ primarily capture short-term market expectations. Overall, these results suggest that both CSR performance and external assurance, in isolation, may be insufficient to generate measurable financial value in the O&G sector, highlighting the importance of industry-specific dynamics in shaping the economic relevance of sustainability practices.

## 6. Conclusion

This study contributes to the ongoing debate on the CSR–financial performance nexus by providing sector-specific evidence from a sample of leading global companies in the O&G industry. The findings indicate that CSR performance does not significantly influence firm valuation, nor does external assurance strengthen this relationship. These findings question the view that sustainability initiatives necessarily translate into measurable financial benefits and emphasise the importance of industry-specific dynamics. From a theoretical perspective, the findings offer important insights into the applicability of dominant CSR frameworks. While Stakeholder Theory predicts that CSR engagement enhances firm value through improved stakeholder relationships and reputational gains, this study suggests that such mechanisms may not operate uniformly across industries. Conversely, the findings are more consistent with Legitimacy Theory, suggesting that in highly scrutinised sectors such as O&G, CSR practices may primarily serve to preserve social acceptance rather than generate direct economic returns. Furthermore, the absence of a significant moderating effect of external assurance suggests that credibility-enhancing mechanisms may not always translate into market rewards, particularly in industries exposed to scepticism and greenwashing concerns. The study also carries practical implications for several stakeholder groups. For managers, the results suggest that CSR initiatives, in isolation, may not deliver immediate financial returns, particularly in high-cost and highly regulated sectors. This highlights the need to move beyond symbolic CSR engagement toward more integrated and strategically aligned sustainability practices that can generate long-term value. For investors, the results underscore the importance of critically evaluating the substance of sustainability initiatives, as CSR performance and assurance may not represent reliable indicators of firm value. For policymakers and regulators, the findings raise questions about the effectiveness of current reporting and assurance frameworks in enhancing transparency and market efficiency, suggesting a need for stronger standardisation and enforcement mechanisms to improve the credibility and comparability of ESG information. Despite its contributions, this study is subject to several limitations. First, the analysis covers a recent but limited time horizon (2022–2023), which may not capture the longer-term financial implications associated with CSR investments and sustainability strategies. Second, the focus on the O&G industry may limit the generalisability of the findings to other sectors with different structural and institutional characteristics. Third, CSR performance is proxied using ESG scores, which, while widely adopted, may not fully reflect the complex nature of CSR activities. Finally, the study does not explicitly account for potential endogeneity issues or omitted variables that may influence both CSR engagement and financial performance. Future research should address these limitations by adopting longer time horizons, incorporating alternative and more granular measures of CSR, and exploring cross-industry and cross-country comparisons. Additional research may also explore the conditions under which external assurance enhances the credibility and value relevance of sustainability disclosures, thereby contributing to a deeper understanding of the evolving role of ESG practices in financial markets and corporate strategy.

**Author Contributions**

Conceptualization, T.I. and A.R.; methodology, A.R.; software, A.R.; validation: A.R. and G.R.; formal analysis, T.I.; investigation, T.I. and A.R.; data curation, A.R. and T.I.; writing—original draft preparation, T.I. and A.R.; writing—review and editing, T.I.; supervision, G.R.; project administration, G.R. and A.R. All authors have read and agreed to the published version of the manuscript.

**Data Availability**

The data used to support the research findings are available from the corresponding author upon request.

**Conflicts of Interest**

The authors declare no conflicts of interest.

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